Computer Architecture

Course Information

- Course Homepage

  New ee-class system
  https://ncueeclass.ncu.edu.tw/

- Instructor: 鄭旭詠

  e-mail : chengsy@csie.ncu.edu.tw

- Teaching Assistants:

陳奕羱 <yiyuan0918@gmail.com>、
王聖允 <yzu1103309@gmail.com>
鄭旭佑 <avery2070527@gmail.com>

Course Information

Textbook

- J. L. Hennessy and D. A. Patterson, Computer Architecture: A Quantitative Approach, sixth Edition, Morgan Kaufmann Publishing Co.

- D. A. Patterson and J. L. Hennessy , Computer Organization & Design: The hardware/software interface, Morgan Kaufmann Publishing Co.

Grading Policy

- 40% Midterm
- 40% Final Examination
- 20% In-class Quiz

* Midterm and Final are closed-book exams.

* Open-book and discussion permitted for all quizzes.

Reference Resources

- David Patterson
  - Electrical Engineering and Computer Sciences, University of California, Berkeley
  - http://www.eecs.berkeley.edu/~pattrsn
  - http://www-inst.eecs.berkeley.edu/~cs252

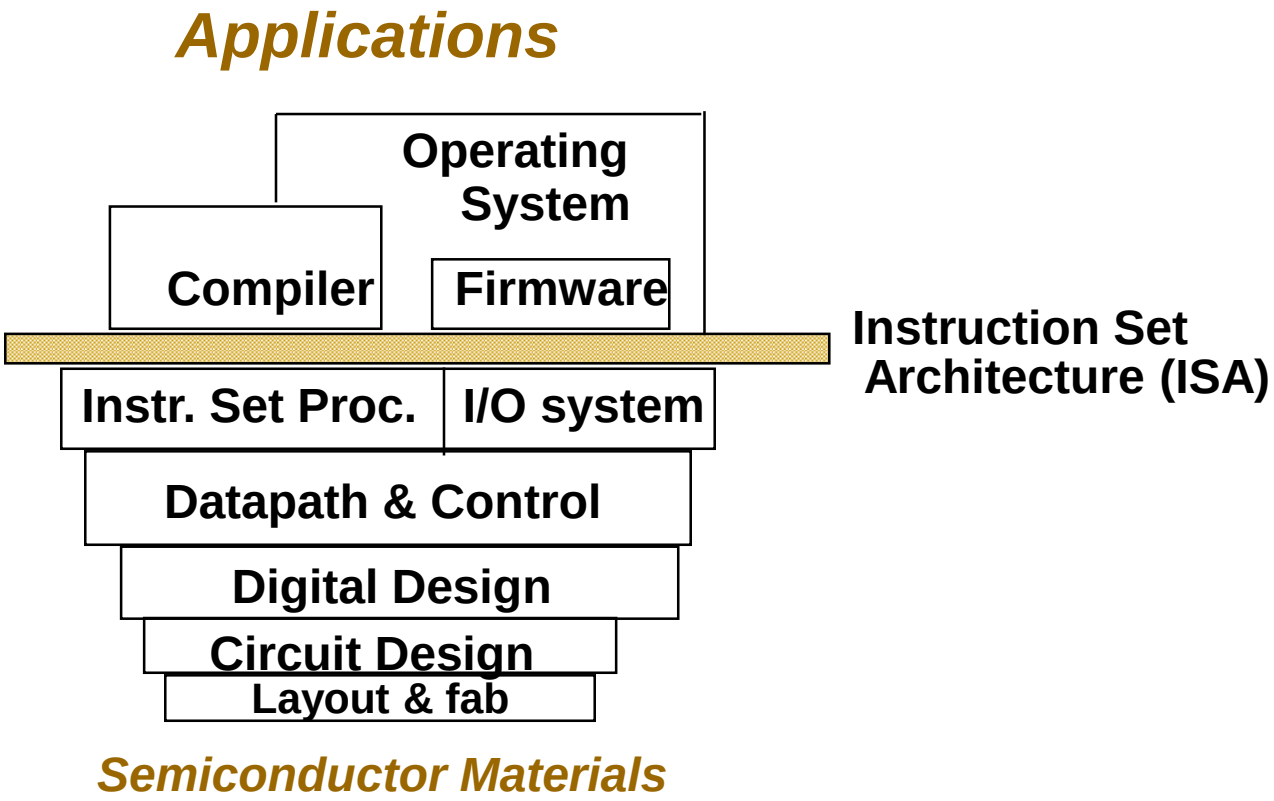
- Shih-Hao Hung, National Taiwan University

  Graduate Computer Architecture

  http://www.csie.ntu.edu.tw/~hungsh/CA/ca.htm

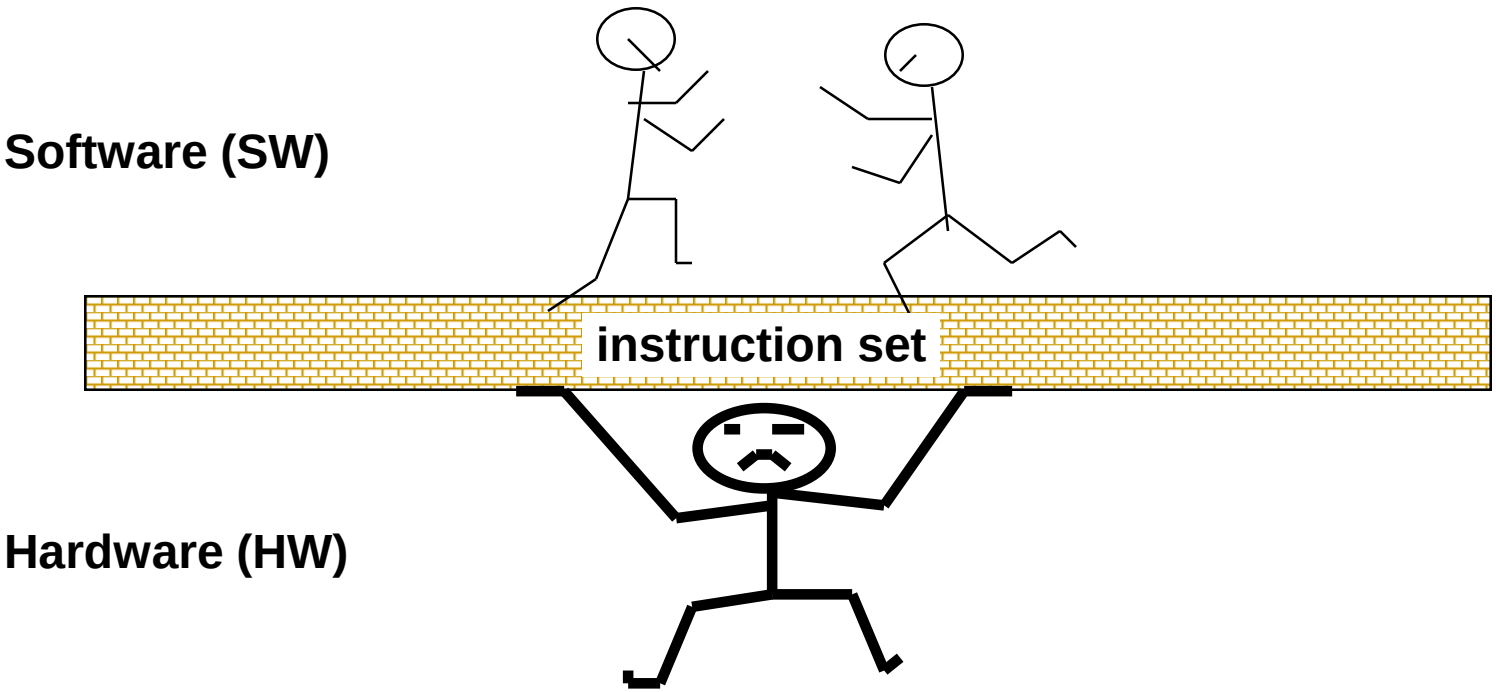
- David E. Culler, UC-Berkeley
  http://www.cs.berkeley.edu/~culler/courses/cs252-s05/

What is "Computer Architecture"?



- Coordination of many *levels of abstraction*
- Under a rapidly changing set of forces
- Design, Measurement, *and* Evaluation

The Instruction Set: a Critical Interface



- Properties of a good abstraction
  - Lasts through many generations (portability)
  - Used in many different ways (generality)
  - Provides convenient functionality to higher levels
  - Permits an efficient implementation at lower levels

Computer Architecture Topics

Input/Output and Storage

Disks, WORM, Tape

RAID

DRAM

Emerging Technologies
Interleaving
Bus protocols

Memory Hierarchy

L2 Cache

Coherence,
Bandwidth,
Latency

L1 Cache

VLSI

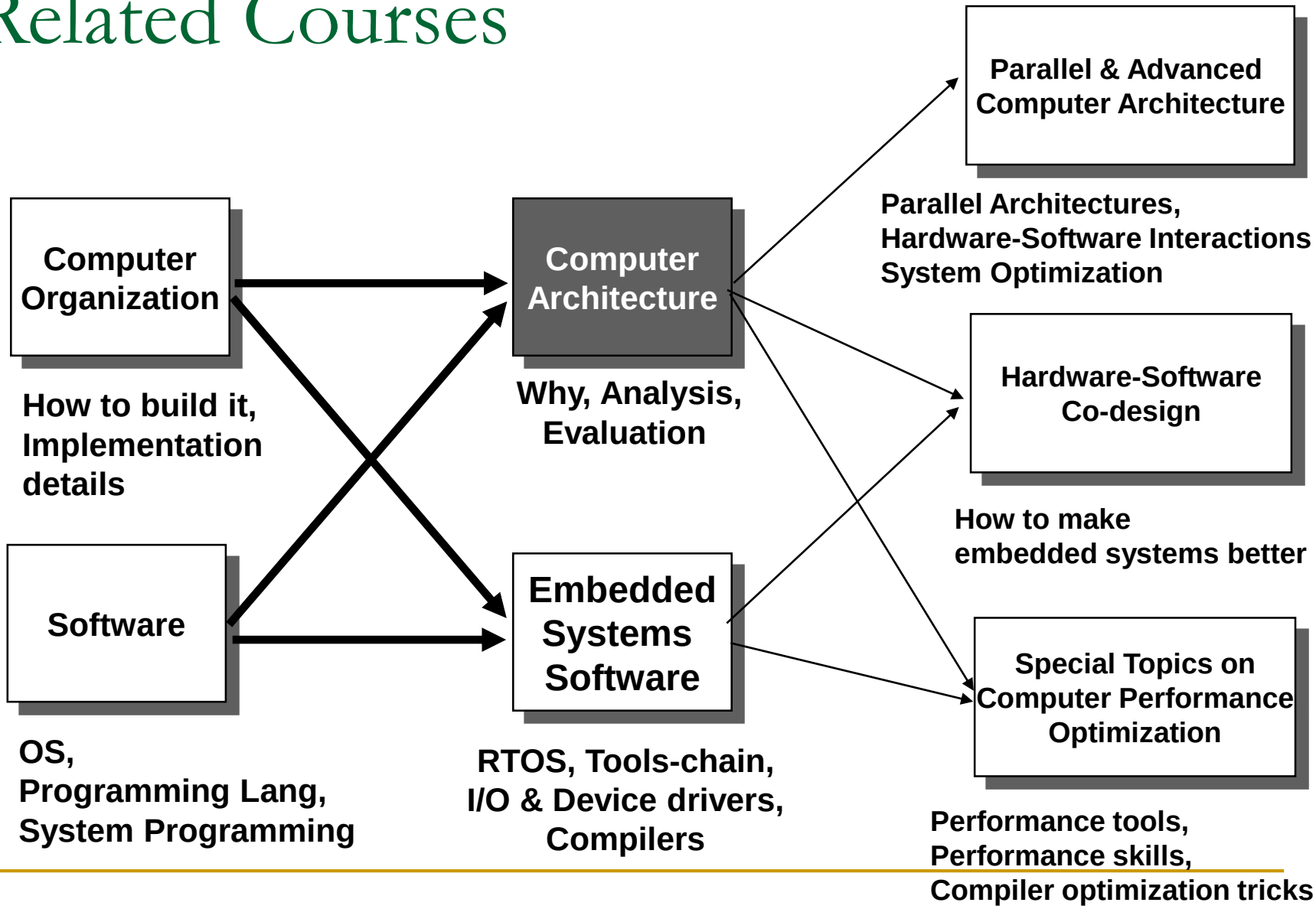
Instruction Set Architecture

Addressing,
Protection,
Exception Handling

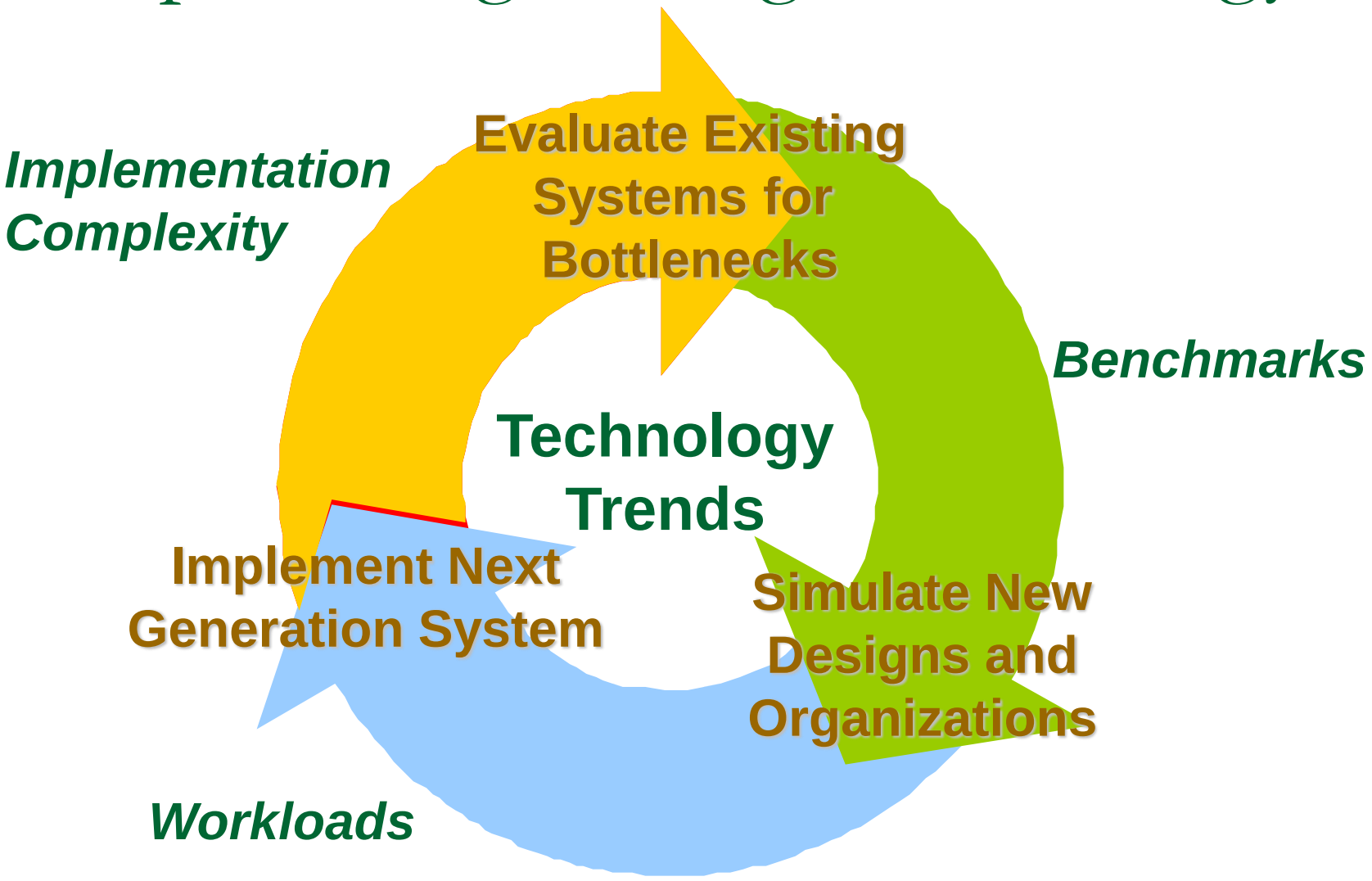
Pipelining, Hazard Resolution,
Superscalar, Reordering,
Prediction, Speculation,
Vector, DSP

Pipelining and Instruction Level Parallelism

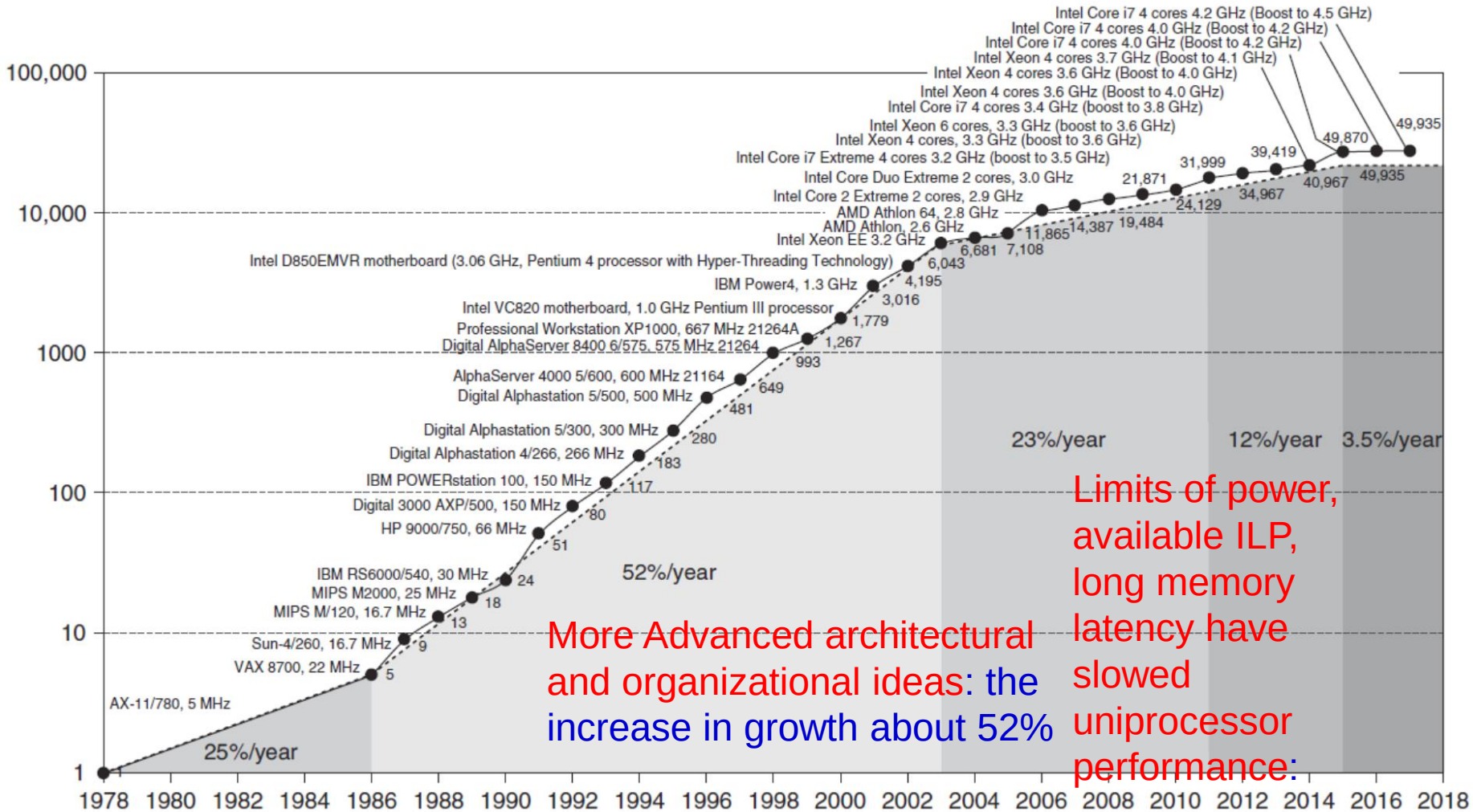
Related Courses



Computer Engineering Methodology



Growth in Process Performance since the mid-1980s



Before 1986:
technology driven:
25% per year

Crossroads: Conventional Wisdom in Comp. Arch

- Old Conventional Wisdom: Power is free, Transistors expensive
- New Conventional Wisdom: "Power wall" Power expensive (Maximum power dissipation of air-cooled chips)

- Old CW: Sufficiently increasing Instruction Level Parallelism via compilers, innovation (Out-of-order, speculation, VLIW, …)
- New CW: "ILP wall" law of diminishing returns on more HW for ILP

- Old CW: Multiplies are slow, Memory access is fast
- New CW: "Memory wall" Memory slow, multiplies fast (200 clock cycles to DRAM memory, 4 clocks for multiply)

- Old CW: Uni-processor performance 2X / 3 yrs
- New CW: multiple "cores"

Crossroads

- In 2004, Intel canceled its high-performance uni-processor projects and joined IBM and Sun
- Declaring that the road to higher performance would be via multiple processor per chip rather than faster uni-processors
- Little Instruction Level Parallelism (ILP) left to exploit efficiently => switch to Thread Level Parallelism (TLP) and Data Level Parallelism (DLP)

- ILP: Compiler and hardware conspire to exploit ILP implicitly without programmer's attention
- TLP and DLP: explicitly parallel, requiring the programmer to write parallel code to gain performance

Attendance & Quiz Regulations

- **1. Attendance Policy**
  - Students should attend physical classes (in-person) on odd or even weeks based on the last digit of their student ID numbers, as specified on the following slide.
  - Other students may join the class online.

- **2. Quiz Submissions**
  - Students attending online can submit their quizzes via the **ee-class system**.
  - **Important:** If you do not meet the criteria for online attendance (i.e., you are supposed to be in class but stayed online), your quiz submission will **not be graded**.

Syllabus

Even/Odd		
	2 / 27	Leave Adjustment for Peace Memorial Day- NO CLASSES
	3 / 6	Course Overview
Odd	3 / 13	Fundamentals of Computer Design
Even	3 / 20	Instruction Set Principles and Pipelining
Odd	3 / 27	Advanced Pipelining and Instruction Level Parallelism (ILP)
	4 / 3	Leave Adjustment for Children's Day- NO CLASSES
Even	4 / 10	ILP, Scoreboard
	4 / 17	Midterm Exam
Odd	4 / 24	Scoreboard, Tomasulo
	5 / 1	Labor Day- NO CLASSES
Even	5 / 8	Branch Prediction
Odd	5 / 15	Branch Prediction, Multi-Processers
Even	5 / 22	Multi-Processers
Odd	5 / 29	Memory Hierarchy
Even	6 / 5	Virtual Memory, Storage
	6 / 12	Final Exam

Student Leave Regulations / Leave Policy

- All leave requests must comply with NCU's student leave regulations.

- Students who are unable to attend semester examinations due to official duties, pregnancy, childbirth, severe illness, bereavement of immediate family members, or other force majeure events must submit a leave. Such applications must be coordinated with the Office of Academic Affairs before a make-up examination can be requested.